# Robust and Calibrated Machine Learning Models for Alzheimer's Disease Risk Prediction

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***Abstract*—Risk prediction of Alzheimer disease (AD) should be accurately and reliably used as clinical decision support. The paper analyses the strength of various machine learning models using structured clinical and cognitive data with special attention to quality of calibration and performance stability as opposed to accuracy. It has been experimentally discovered that ensemble models give more accurate predictions about probabilities in data splits. On the basis of discrimination, calibration and stability, models were evaluated and a systematic preprocessing was necessary to ensure a fair evaluation and reproducible evaluation.**

***Index Terms*—Alzheimer's disease prediction, Machine learning, Model robustness, Probability calibration, Stability analysis**

## I. Introduction

Alzheimer disease (AD) is the most common cause of dementia and presents an ever-increasing health challenge in the world, especially where the elderly are concerned. Research and studies have provided the ideal answer to this given that timely clinical intervention and effective management in the disease require accurate and early identification of individuals that are at the relatively high risk. Although the neuroimaging and biomarker approaches offer good diagnostic data, the technique is quite expensive, invasive, or not feasible as a large scale screening tool. Consequently, there has been growing attention in using routinely sampled clinical, cognitive and lifestyle information to predict AD risk [1].

The initial research in AD prediction was mainly based on the conventional statistical techniques, including logistic regression, which are highly interpretable yet might not be able to depict the sophisticated nonlinear connection among the heterogeneous clinical characteristics [2]. As machine learning methods have been developed, increasingly expressive models have been shown to exhibit enhanced predictive accuracy on structured clinical data, such as support vector machine, ensemble tree models and gradient boosting [3], [4]. Nevertheless, much of the existing literature concentrates much on discrimination measures including accuracy or AUC, and not on the probability calibration and performance stability, which are essential in clinical decision support [5].

This paper systematically reviews machine learning models for Alzheimer's disease prediction using structured clinical and cognitive features, with emphasis on probability calibration, performance stability across repeated data splits, and feature importance for interpretability. The remainder of the paper is organized as follows: Section II reviews related work, Section III describes the methodology, Section IV presents experimental results, and Section V concludes with future research directions.

## II. Related Work

Initial research on predicting AD was mainly based on conventional statistical and machine learning systems with structured clinical and cognitive measures. The use of logistic regression and other linear classifiers was largely accepted because of its simple interpretation and implementation as well as inability to model all nonlinear relationships [2]. Further developments presented support vector machines and the use of ensemble models, which were shown to be better at classification as they were capable of capturing more feature interactions [6].

As more clinical data is available, tree-based ensemble models and gradient boosting methods have gained popularity in predicting the AD that are highly discriminative in a variety of cohorts [4]. Nevertheless, a vast majority of available research places more emphasis on accuracy or AUC, leaving a relatively simple examination of probability calibration and split robustness. More recent efforts have focused on the need to assess the quality of calibration, and studying the stability of the model as the basis of efficient clinical decision support with the understanding that high discrimination is not alone suitable in practical implementation [5].

## III. Methodology

### *A. Dataset*

In this research, a publicly available dataset of clinical diseases of Alzheimer disease, which was stored on Kaggle, is used, health records of 2, 149 people aged 60-90 years [7]. Each of the subjects will be characterised by the structured variables that will include demographics, lifestyle factors, medical history, clinical measurements, cognitive and functional assessment. Cognitive status is measured by the use of such standardized assessment tools as the Mini-Mental status examination (MMSE) and activities of daily living (ADL). The dichotomous diagnosis subject is used to determine the presence or absence of the AD. The dataset is divided into an independent test set of 430 subjects and a training set of 1, 719 subjects in an 80/20 splits.

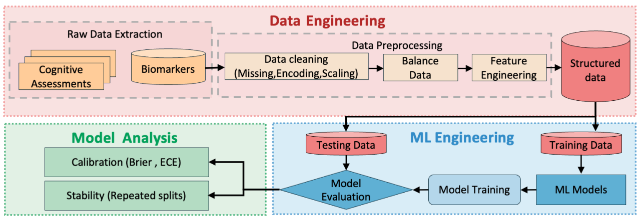


Fig. 1: Overview of the robustness-oriented machine learning evaluation pipeline for AD prediction

Fig. 1 illustrates the evaluation pipeline, including data preprocessing, class balancing, feature selection, and model training, with performance assessed using discrimination metrics alongside calibration and stability analyses for AD prediction.

### *B. Data preprocessing*

To ensure consistency across all models, the following preprocessing steps were applied:

**Remove non-informative variables:** Identifier attributes, including PatientID and DoctorInCharge, were removed as they do not contribute to prediction.

**Duplicate handling:** Duplicate columns were detected and eliminated to avoid redundancy.

**Missing value imputation:** Missing values were handled using median imputation, which is robust to outliers and suitable for clinical data.

**Feature scaling:** Continuous variables were standardized to zero mean and unit variance.

**Categorical encoding:** Binary variables were encoded using label encoding, while nominal variables (e.g., ethnicity and education level) were transformed via one-hot encoding. This unified preprocessing pipeline ensures fair and reliable comparison across different models.

### *C. Balance data*

The training set exhibits moderate class imbalance, with an AD-positive rate of 0.354. Class imbalance is common in clinical prediction tasks and may bias learning toward majority-class patterns. While prior studies frequently apply resampling methods such as SMOTE, synthetic samples may distort the original data distribution and adversely affect probability calibration. To avoid these issues, a cost-sensitive learning strategy with inverse-frequency class weighting was adopted to preserve the original data structure while emphasizing minority-class signals.

### *D. Feature Selection*

The processed data has 39 variables in structured clinical and cognitive forms that can cause redundancy and the risk of overfitting. They did not embrace dimensionality reduction techniques like PCA, which transform original variables into latent factors and mask clinical interpretation. In their place, an embedded feature selection method, which relied on the L1-regularized logistic regression, was utilized so as to encourage sparsity by scaling down tiny coefficients to zero. The absolute coefficients were sorted and top 10 predictors were taken.

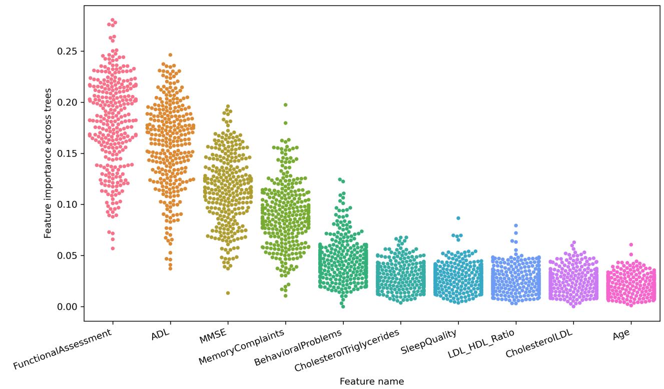


Fig. 2: Distribution of feature importance scores for the top 10 selected clinical and cognitive features

These features as seen in Fig. 2 are always of high importance in the repeated training of the models which confirms the consistency and explanation of the chosen feature set. Moreover, LDL/HDL ratio was calculated directly to include the cardiovascular risk factors that were pertinent to AD.

### *E. Model Architectures*

To fully estimate the ability of structural clinical data to predict the risk of AD, five representative machine learning models and using the linear, kernel-based, ensemble, and deep tabular paradigms were utilized. The comparison between all models was made to be fair, as all models were trained on an equal preprocessed set of features.

#### *1) Logistic Regression (LR)*

Logistic regression is an interpretable binary clinical prediction base regression. With an input feature vector $\mathbf{x} \in \mathbb{R}^d$, the predicted probability of Alzheimer's disease can be defined as

$$P(y = 1 \mid \mathbf{x}) = \frac{1}{1 + \exp\left(-(\mathbf{w}^\top \mathbf{x} + b)\right)}, \tag{1}$$

where $\mathbf{w}$ is the coefficients of the model and $b$ is the bias. In the effort of alleviating the problem of imbalance between the classes, the loss function was weighted by the classes, which enhanced the sensitivity to the minority-class patterns.

#### *2) Support Vector Machine (SVM)*

A radial basis function (RBF) kernel support vector machine kernel was used, as this was found to be effective at capturing nonlinear combinations of features. The decision function can be described as giving follows

$$f(\mathbf{x}) = \sum_{i=1}^{N} \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b, \tag{2}$$

where $K(\cdot, \cdot)$ denotes the RBF kernel. Probabilistic outputs were obtained via Platt scaling to enable calibration analysis.

#### *3) Random Forest (RF)*

The random Forest is an ensemble learning algorithm which consists of several decision trees that are trained by bootstrap aggregation. The result is the average of the individual tree outputs:

$$\hat{y} = \frac{1}{T} \sum_{t=1}^{T} h_t(\mathbf{x}), \tag{3}$$

where $h_t(\cdot)$ represents the $t$-th decision tree. This model is effective in terms of capturing feature interaction with the noise-resistance.

*4) Extreme Gradient Boosting (XGBoost)*

XGBoost constructs an additive ensemble of decision trees by minimizing a regularized objective function:

$$\mathcal{L} = \sum_i \ell(y_i, \hat{y}_i) + \sum_k \Omega(f_k), \tag{4}$$

where $f_k$ denotes the $k$-th tree and $\Omega(\cdot)$ is a regularization term fused to the model. The training was done using a scaled positive-class weight to solve class imbalance.

*5) TabNet*

TabNet is a sequential attention, sequential attention, and tabular data deep neural architecture. The feature selection mask is learned at each decision step $t$, a feature selection mask is learned to sparsely select informative features:

$$\mathbf{M}^{(t)} = \text{Sparsemax}\left(\mathbf{A}^{(t)}\right), \qquad \mathbf{x}^{(t)} = \mathbf{M}^{(t)} \odot \mathbf{x}, \tag{5}$$

where $\mathbf{A}^{(t)}$ denotes the attention logits at step $t$, $\mathbf{M}^{(t)}$ is the resulting sparse mask, and $\odot$ represents element-wise multiplication. The last prediction consolidates representations over several decision steps such that it is possible to select features automatically and predict better.

### *F. Calibration and Stability Evaluation Standard*

To assess probabilistic reliability beyond discrimination, calibration and stability analyses were conducted. Calibration evaluates the agreement between predicted probabilities and observed outcomes, assessed using reliability diagrams and quantified by the Brier score and Expected Calibration Error (ECE).

The Brier score measures the mean squared difference between predicted probabilities and true labels:

$$\text{Brier} = \frac{1}{N}\sum_{i=1}^{N}(p_i - y_i)^2, \tag{6}$$

where $p_i$ denotes the predicted probability for the $i$-th sample and $y_i \in \{0, 1\}$ is the corresponding ground-truth label.

ECE was computed by partitioning predictions into $M$ confidence bins and calculating the weighted average difference between accuracy and confidence:

$$\text{ECE} = \sum_{m=1}^{M}\frac{|B_m|}{N}\left|\text{acc}(B_m) - \text{conf}(B_m)\right|, \tag{7}$$

where $B_m$ denotes the set of samples assigned to the $m$-th confidence bin, $\text{acc}(B_m)$ represents the empirical accuracy within the bin, and $\text{conf}(B_m)$ denotes the average predicted confidence.

The test of robustness was made by repeating stratified train-test splits. Second split retraining of models and evaluation metric distribution analysis were done to quantify data variation sensitivity of models.

## IV. Results and Discussion

Five standard metrics were used in the evaluation of the models on the test set with precision, recall, F1-score, accuracy, and AUC being those metrics Fig. 3 shows the ROC curves of all the evaluated models, whereas Table I summarizes appropriate quantitative results.

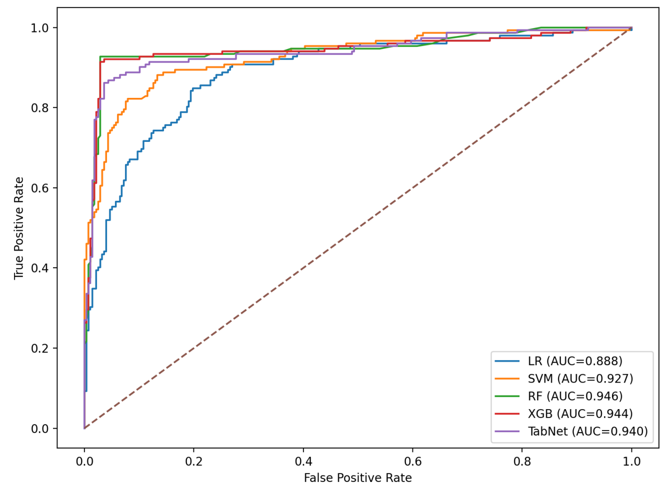


Fig. 3: ROC curves

TABLE I: PERFORMANCE COMPARISON

| Model | Accuracy (%) | F1 (%) | Precision (%) | Recall (%) |
|---|---|---|---|---|
| LR | 81.86 | 76.79 | 70.11 | 84.87 |
| SVM | 86.74 | 81.55 | 80.25 | 82.89 |
| RF | **95.35** | **93.33** | **94.59** | **92.11** |
| XGBoost | 94.88 | 92.72 | 93.33 | **92.11** |
| TabNet | 90.23 | 86.54 | 84.38 | 88.82 |

### *A. Performance Comparison*

Random Forest was found to perform the best on the overall evaluation, with the highest AUC and classification accuracy, as well as the highest F1-score, that is, with a well-balanced score between precision and recall. XGBoost came in right behind with similar levels of discriminative performance, and ascertained the suitability of ensemble-based learning to clinical risks prediction. TabNet showed that its performance is competitive but somewhat lower than that of linear models and is lower than tree-based ensembles. Conversely, the Support Vector Machine and Logistic Regression had lower discriminative strength and average accuracy with the latter having the lowest performance due to the inherent nature of linear decision-making important to capture potential clinically feature interactions.

### *B. Calibration Analysis*

The reliability diagrams and the Brier score and expected calibration error were used to evaluate the calibration performance as presented in Fig. 4 and Table II.

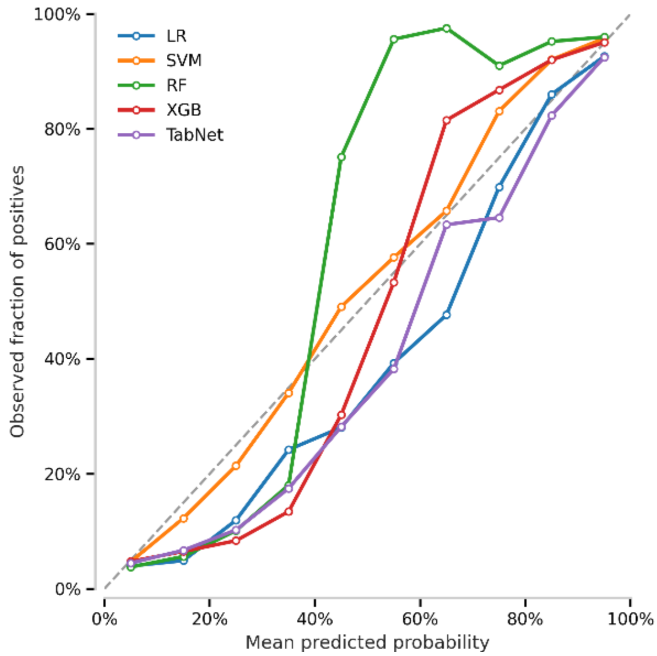

Fig. 4: Reliability diagrams for model calibration

TABLE II: CALIBRATION PERFORMANCE ACROSS MODELS

| Model | Brier Score (mean ± std) | ECE (mean ± std) |
|---|---|---|
| LR | 0.1192 ± 0.0096 | 0.0776 ± 0.0067 |
| SVM | 0.0912 ± 0.0086 | 0.0443 ± 0.0072 |
| RF | 0.0545 ± 0.0079 | 0.0634 ± 0.0096 |
| XGBoost | **0.0501 ± 0.0081** | **0.0373 ± 0.0069** |
| TabNet | 0.0770 ± 0.0133 | 0.0541 ± 0.0117 |

Among all evaluated models, XGBoost showed the highest calibration results with the lowest Brier score (0.0501 ± 0.0081) and ECE (0.0373 ± 0.0069). Its reliability curve almost traced the ideal diagonal in most probability bins, meaning correct and consistent probabilistic estimates. On the contrary, the Logistic Regression gave too conservative estimates of probabilities and exhibited the poorest calibration. The Support Vector machine had moderate calibration as well as slight underconfidence amongst the high probabilities. Random Forest was overconfident in the middle probability range, but TabNet was intermediate in calibration with deviations in mid-to-high probability intervals.

The evaluation of model stability was done by measuring the variance of the AUC, accuracy and F1-score through repeated stratified data splits. XGBoost had the best overall stability with a mean AUC of 0.9486, which has a low standard deviation of 0.0116, combined with a mean F1-score of 0.9258 0.0138. Its accuracy was also relatively constant (0.9486 ± 0.0089). The maximum mean AUC (0.9530) and F1-score (0.9299) were obtained with the random Forest though with an even greater variability. In contrast, LR and SVM exhibited significantly more performance dispersion, in terms of F1-score, in particular. TabNet was found to be intermediate and every mean performance was moderate with an increase in variability. On the whole, these findings provide evidence that XGBoost is the most appropriate to work with in terms of balanced and reliable stability profile.

## V. Conclusion

This study evaluates five machine learning models as clinical risk predictors using structured cognitive and clinical features under a unified and reproducible pipeline. Results show that ensemble-based tree models outperform linear and margin-based approaches. Random Forest achieves the strongest discrimination, while XGBoost demonstrates superior calibration and robustness with comparable discrimination, consistent with prior findings on probability estimation [5]. In contrast, Logistic Regression and Support Vector Machine exhibit weaker predictive power and reliability. TabNet demonstrates competitive performance but remains sensitive to data variability, suggesting higher data demands for stable optimization.

Importantly, this study emphasizes that high discriminative measures do not in themselves suffice to be deployed in clinical settings. The calibration and stability results indicated that there were significant divergences across the models that indicated that reliable probability estimation and the ability to resist sampling variation would be vital when making reliable clinical decisions [8].

Looking forward, future work might be possible to improve model reliability through further development of future work, including explicit calibration methods [9], testing models on external cohorts [4], and use of explainability features like SHAP [10]. Overall, the results identify XGBoost as the most balanced approach and highlight the need for holistic evaluation frameworks that emphasize reliability, calibration, and stability in clinical risk prediction.

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